Faculty Self-Study

CHE 455 – Nanoscale Systems in Chemical Engineering

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This assessment reflects on the presentations delivered by students in CHE 455: Nanoscale Systems in Chemical Engineering, where they explored Scanning Tunneling Microscopy (STM) as a tool for understanding matter at the atomic scale. The goal was to evaluate their technical understanding and their ability to communicate complex scientific ideas.

Students were asked to present on STM, covering its physical principles, instrumentation, and applications. In particular, they were expected to explain quantum tunneling, describe how STM achieves atomic resolution, and connect theory with real-world applications such as surface imaging and electronic structure analysis.

What stood out most was the clarity with which students explained STM concepts. They made good use of diagrams, maintained a logical flow, and demonstrated confidence in presenting technical material. Importantly, they were able to connect abstract ideas, like quantum tunneling, to tangible outcomes such as atomic-scale imaging.

Based on the assessment results, the course is clearly successful in helping students reach a solid level of proficiency, particularly in articulating ideas, presenting professionally, and following structure. However, the variation observed in organization, use of visuals, and writing quality points to a few targeted opportunities to strengthen the course.

Future presentations could benefit from more emphasis on the underlying physics for instance, distinguishing between topographic and electronic contributions to STM images, or discussing the role of local density of states (LDOS) in greater detail. There is also an opportunity to encourage more critical thinking by asking students to analyze limitations or interpret experimental scenarios.

Another area for improvement is the effective use of figures and diagrams. Students are already including visuals, but they are not always using them as analytical tools. I would introduce a dedicated activity where students must interpret a figure (e.g., STM image or LDOS plot) and explain its physical interpretation.

The results also suggest room to strengthen professional writing and technical polish. Rather than treating this as an expectation alone, I would consider

integrating lightweight feedback checkpoints. For example, a draft slide review or a short-written summary before the final presentation. This would allow students to correct issues in grammar, tone, and clarity earlier in the process.